Krishnaa **Vadivel** | Computational Physics PhD | First-Principles Materials | HPC | Scientific ML

516-853-5663 • srikrishnaa.careers@gmail.com • sri-krishnaa-ece-phd
Website | GitHub: krishnaa423 | LeetCode: krishnaa42342 | Docker Hub: krishnaa42342

Profile

Computational physics PhD researcher with training across electrical engineering, condensed matter, semiconductor physics, numerical methods, and scientific software. Current work centers on first-principles calculations of self-trapped excitons and excitonic polarons, many-body formulations of electron-phonon coupled systems, and scalable implementations for national-lab HPC platforms.

Research Experience

Yale University, Electrical Engineering
Research Assistant / PhD Researcher *2021–present*

- Develop first-principles research workflows for self-trapped excitons, excitonic polarons, and related excited-state phenomena in materials, linking DFT, DFPT, GW/BSE-style ideas, and effective Hamiltonian viewpoints.
- Study many-body and field-theoretic formulations of coupled electron, exciton, and phonon systems, including coherent-state and path-integral approaches for time-dependent and self-localized states.
- Implement distributed scientific computing pipelines in Python and C++ with MPI, OpenMP, PETSc, and SLEPc for large linear-algebra and simulation workloads.
- Port and optimize computational workloads for accelerator-enabled HPC environments using CUDA and ROCm, with experience running on Frontier, Perlmutter, and other leadership-class clusters.
- Investigate machine-learning accelerators for scientific computing, including equivariant graph neural networks for molecular-property prediction and ML-assisted materials workflows.
- Co-authored a 2025 *Journal of the American Chemical Society* paper and presented APS talks in 2024, 2025, and 2026 on self-trapped exciton formation and related excited-state materials physics.

Professional Experience

Probe Technologies Inc. / Weatherford Wireline Services
Software and Embedded Systems Engineer *2019–2021*

- Built CUDA-based data-processing pipelines and custom GPU-enabled engineering tooling for large acoustic waveform datasets generated by downhole logging tools.
- Developed embedded software and digital logic for sensor logging and tool control using Lattice FPGA, PIC32, dsPIC, STM32, and C8051-based systems.
- Supported hardware-oriented engineering tasks including circuit-level debugging, oscilloscope-based validation, field testing, and targeted PCB modifications in Altium.
- Contributed host-side engineering software with C#, ASP-based services, and Microsoft SQL-backed database workflows.

FindLight
Photonics Technical Writer *2018–2019*

- Produced technical articles and product-facing photonics content covering lasers, optics, and optoelectronic components for a specialized engineering readership.

Selected Projects

Distributed First-Principles Exciton-Phonon Simulation

- Developed computational workflows for first-principles excited-state and exciton-phonon calculations, with an emphasis on self-trapped excitons, scalable many-body formulations, and reusable HPC pipelines.

Semiconductor Device Simulation

- Built numerical treatments of semiconductor transport and electrostatics, including drift-diffusion and pn-junction-style device equations, to connect condensed-matter models with engineering-scale simulation.

Equivariant Graph Neural Networks for Molecular Systems

- Implemented graph-based and symmetry-aware ML models for molecular-property prediction and explored their use as accelerators for scientific workflows.

Agentic Scientific Input Generation

- Built retrieval-driven tooling that reads software documentation and research papers to generate structured computational-science inputs and starter workflows for physics and chemistry codes.

Integrated Photonics and FDTD Modeling

- Simulated photonic structures including band-pass and resonator-style devices with FDTD tooling such as Meep.

Education

Yale University
PhD, Electrical Engineering *2021–present*
Ongoing doctoral research in computational materials, many-body theory, and scientific computing.

Yale University
MPhil, Electrical Engineering *2023*

Yale University
MS, Electrical Engineering *2023*

Cornell University
MEng, Electrical and Computer Engineering *2017–2018*
Included work in stochastic computing, device-oriented EE topics, and advanced design coursework.

Cornell University
BS, Electrical and Computer Engineering *2013–2017*
Included embedded systems, robotics, and core mathematics, physics, and computing foundations.

Selected Publications

2025: de Paula, A. M., Li, S., Hou, B., Vadivel, S., Teles-Ferreira, D. C., Iudica, A., Kabacinski, P., Hosseini, H., McArthur, J., Cerullo, G., et al. (2025). Time-domain observation of ultrafast self-trapped exciton formation in lead-free double halide perovskites. *Journal of the American Chemical Society, 147*(32), 28923–28931.

Selected Conference Presentations

2026: Vadivel, S., Grande, R. D., Strubbe, D. A., & Qiu, D. Y. (2026). First-principles study of exciton-polarons and self-trapped excitons. *APS Global Physics Summit 2026*.

2025: Vadivel, S., Qiu, D. Y., Grande, R. D., & Strubbe, D. A. (2025). First-principles study of self-trapped exciton formation in double perovskites. *APS Global Physics Summit 2025*.

2024: Vadivel, S., & Qiu, D. (2024). First-principles study of self-trapped exciton formation in double perovskites using excited state forces. *APS March Meeting Abstracts 2024*, F01–007.

Selected Coursework

Mathematics: Differential Geometry; Abstract Algebra; Honors Mathematical Analysis II

Computer Science: Theoretical Computer Science; Probabilistic Machine Learning

Physics: Graduate Quantum Mechanics I–II; Solid State Physics I–II; Many-Body Quantum Physics

Electrical Engineering: Semiconductor Physics; Thin Film Science; Lasers and Optoelectronics; Integrated Photonics; VLSI Design; Analog VLSI Design; Mathematics of Signals and Systems; Advanced Embedded Systems

Technical Skills

Languages: Python, C, C++, CUDA, JavaScript, Bash, PowerShell

Scientific Computing: MPI, OpenMP, PETSc, SLEPc, NumPy, pandas, SciPy, SymPy, matplotlib, mpi4py, petsc4py, slepc4py

Machine Learning: PyTorch, PyTorch Geometric, scikit-learn, equivariant graph neural networks

Systems: Linux command line, macOS, Windows, CMake, Docker, Docker Compose, Kubernetes, gcloud

Hardware: Lattice FPGA, PIC32, dsPIC, STM32, C8051, PCB bring-up and instrumentation-based debugging

HPC Platforms: Frontier, Frontera, Perlmutter, and related national-lab or university HPC environments